

## ELEMENT B

When exploring solutions to problems involving water accumulation on sidewalks or walkways, several design attempts have been used in the past to manage drainage and prevent flooding. Each of these designs offers valuable insight into what works, what fails, and why certain approaches may not fully meet the needs of environments such as school campuses. By examining these prior design attempts, we can better understand the limitations of existing systems and identify opportunities for improvement.

One common approach is the under-sidewalk drain system, which uses pipes installed beneath the sidewalk to carry water away from the surface. Small openings in the sidewalk allow rainwater to enter the pipes, helping reduce pooling during storms. According to AlSyed Construction Company (2025), these systems are intended to prevent flooding by directing water underground and away from pedestrian areas. However, this design often fails during heavy rainfall because the drains may be too small or improperly placed, causing water to continue collecting on the surface. Additionally, debris such as dirt and leaves can clog the openings, reducing the system's effectiveness over time. These issues show that while the concept is useful, the design does not consistently meet drainage requirements in real-world conditions.

Another widely used solution is the rain garden, a landscaped area designed to absorb runoff from roofs, sidewalks, and streets. Rain gardens temporarily hold water and allow it to soak into the ground, typically drying out within one to two days to prevent mosquito breeding (Live in Your Backyard, 2024). Although rain gardens are environmentally friendly and visually appealing, they are not always reliable during severe storms. If the garden is too small or not maintained properly, it can overflow, failing to prevent water from spreading onto sidewalks. This makes rain gardens helpful but not fully dependable for high-volume drainage needs.

A simpler method involves constructing sloped sidewalks, which use gravity to guide water toward drains or grassy areas. Valor Environmental (2025) explains that sloping surfaces can reduce water buildup by encouraging natural flow. However, this approach has limitations. If the slope is too shallow, uneven, or poorly designed, water can still collect in low spots. During intense storms, even well-sloped sidewalks may not drain quickly enough, allowing puddles to form. This shows that slope alone cannot guarantee effective drainage in all conditions.

Another prior design attempt is the French drain, a trench filled with gravel and a perforated pipe that redirects water away from sidewalks. French drains are commonly used to manage groundwater and surface runoff. Attic Pros (2026) notes that these systems can be effective when installed correctly, but they are prone to clogging if soil or debris enters the pipe. If the trench is not built at the proper angle, water may not flow through it efficiently, causing backups or blockages. These weaknesses limit the long-term reliability of French drains, especially in areas with heavy rainfall or poor soil conditions.

Finally, permeable pavement has been used as a modern solution to sidewalk flooding. This type of pavement allows water to pass through the surface and into the ground below, reducing runoff. Family Handyman (2023) highlights that permeable pavement can significantly reduce standing water. However, over time, dirt and sediment can clog the pores in the pavement, preventing water from soaking through. In regions with very heavy rain or compacted soil, permeable pavement may not drain quickly enough to prevent pooling, making it less effective in demanding environments.

Across all five prior design attempts, a common theme emerges: each solution works under certain conditions but fails when drainage demands exceed the system's capacity or when maintenance is neglected. Clogging, improper installation, limited capacity, and environmental factors all contribute to the shortcomings of these designs. Understanding these limitations is essential for developing a more reliable and sustainable solution that can effectively manage water in school environments and prevent hazards such as slipping, flooding, and structural damage.